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A Dual Approach to Profitability and Sustainability: AI-Powered Pricing and Emissions Control in Textiles

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ABSTRACT Energy management in industrial manufacturing is increasingly driven by decarbonization requirements and cost optimization. This study proposes a digital twin framework that integrates photovoltaic (PV) energy forecasting, carbon footprint accounting, and AI-based dynamic pricing for a vertically integrated textile manufacturer, ATLAS Denim Co. in Adana, Türkiye. Using historical meteorological data and on-site PV generation records, three machine learning algorithms—Random Forest (RF), Extreme Gradient Boosting (XGBoost), and Long Short-Term Memory (LSTM)—are trained to forecast solar energy output. The RF model achieves the highest accuracy, with a coefficient of determination (R^2) of 0.993, and is used to estimate PV generation for the January–May 2025 period. The forecasts are combined with production data to compute energy use, carbon emissions, and energy cost per meter of textile output. A carbon-aware pricing model then links product prices to energy mix and emission intensity, enabling the formulation of sustainability-adjusted price offers. Over the five-month horizon, the framework yields an estimated 55.18 MWh of PV generation and a carbon saving of 23.9 tCO_2 compared to grid-only electricity. The results demonstrate that integrating machine learning (ML)-based forecasting and digital twin concepts can support climate-conscious and economically efficient decision-making in textile manufacturing.

INDEX TERMS Digital twin, solar energy forecasting, machine learning, carbon footprint, dynamic pricing, textile manufacturing

I. INTRODUCTION

The industrial sector is responsible for approximately one-third of global final energy use and carbon dioxide (CO_2) emissions, making it a central focus in the global transition toward sustainable energy systems [1]. Among industrial branches, the textile industry stands out for its high energy intensity, complex production processes, and substantial environmental footprint. From spinning and weaving to dyeing and finishing, textile manufacturing demands large volumes of electricity and thermal energy, much of which is derived from fossil fuels [2]. Consequently, the sector faces mounting pressure to adopt cleaner, more efficient production strategies in line with global decarbonization targets and consumer expectations for sustainable products.

In recent years, renewable energy integration into industrial operations has gained considerable momentum, particularly through photovoltaic (PV) systems, due to their modularity, scalability, and falling costs. However, integrating such systems into real-world industrial settings introduces challenges related to intermittency, predictability, and operational optimization. Traditional control systems and static scheduling approaches often lack the flexibility needed to adapt to dynamic production conditions and variable energy availability [3]. As a result, industries are turning to more intelligent, data-driven solutions.

One promising technological paradigm that addresses these challenges is the digital twin—a virtual representation of a physical asset or process that enables real-time monitoring, simulation, and optimization [4]. Originally popularized in aerospace and manufacturing, digital twin systems are now increasingly applied in energy management scenarios, where they offer significant benefits in terms of predictive control, resource optimization, and sustainability analytics [5, 6]. For instance, Schneider Electric's Le Vaudreuil plant in France reported a 25% reduction in energy use and 25% lower CO_2

emissions following the deployment of a digital twin for facility optimization [7]. Similarly, textile-specific implementations have demonstrated promising results: a study by Yılmaz, et al. [8] showed that using digital twin technology in dyeing and finishing processes led to over 12% energy savings and significant reductions in greenhouse gas emissions, especially in small and medium-sized enterprises (SMEs).

Despite these successes, most existing studies focus either on large-scale facilities or isolated processes rather than holistic, factory-wide integration in real-time environments. There is a noticeable lack of applied research on data-centric digital twins in small-scale textile plants, where resource constraints, variability in production schedules, and lack of process standardization hinder the deployment of such advanced systems. Moreover, the integration of artificial intelligence (AI) into digital twins for forecasting renewable energy production and linking these forecasts to operational decisions, such as pricing and emissions reporting, remains an underexplored area.

In the study by Park, et al. [9], the feasibility of a solar hydrail system powered by solar and hydrogen energy that can replace diesel locomotives in South Korea was evaluated with digital twin technology. In a 30-year simulation based on Gangwon-do province data, it was shown that the use of solar hydrail reduced the net present cost by approximately 59% and reduced annual harmful emissions by 300 tons.

In the study of Castilla, et al. [10], a digital twin was proposed for a flat plate solar collector field used to reduce carbon emissions in the transformation to zero-energy buildings. This digital twin, created with an Artificial Neural Network (ANN)-based prediction model, was developed as an alternative to models based on physical equations in systems where nonlinear behaviors are dominant and was validated with one year of operation data. The model was found suitable for system performance prediction, data analysis, and control; it also provided an innovative framework for real-time decision making and sustainable operation by integrating with web-based user interface and data collection technologies.

In the study by Mohamad Zaidi, et al. [11], digital twin technology was used to reduce high residential energy consumption and carbon emissions caused by increasing urbanization in the tropical climate of Malaysia. The digital twin model developed for Bertam City (Penang) was created with the iCL-iCD energy simulation software and it was determined that the electricity consumption of residential buildings for cooling purposes constituted 36% of the city's total energy use. In the initial scenario, residential energy consumption was calculated as 607 GWh and carbon emissions as 314,736 tCO_2e; it was revealed that 37.3% savings could be achieved in these values with the implemented energy efficiency measures.

Mingorance, et al. [12] aims to develop a digital twin–based methodology for real-time optimization of industrial operations and maintenance, integrating reinforcement learning and neural networks to improve efficiency, reduce costs and carbon emissions, and enhance responsiveness to operational disturbances. The approach demonstrates measurable benefits, including a 5% reduction in cost per tone, 5% lower CO_2 emissions, and a 30% improvement in demand adjustment.

In the study of Zhao, et al. [13], improvement scenarios for the transformation of existing buildings into zero-carbon emission within the scope of combating climate change were examined with a digital twin-based assessment framework. A 3D visualization model of the existing building was created with the integration of point cloud data and Building Information Modeling (BIM), and different improvement scenarios were simulated in this virtual environment. Each scenario was evaluated according to criteria such as energy consumption, carbon emission, predicted mean vote, and percentage dissatisfaction. The results showed that the selected zero-carbon improvement scenario reduced energy use by 13.95%, increased solar energy production by 27.80%, and reduced annual carbon emissions by 18,106.32 $kgCO_2$.

In the study of Vilakazi and van Zyl [14], digital twin technology was used to estimate the energy production of a solar PV power plant belonging to the University of Johannesburg in South Africa. Since real measured solar radiation data was not available, Long Short-Term Memory (LSTM) and Recurrent Neural Network (RNN) deep learning algorithms were used to estimate synthetic meteorological data generated by the digital twin. In the performance evaluation, it was determined that the LSTM model was the most successful estimator according to Root Mean Squared Error (RMSE), MSE, and Mean Absolute Error (MAE) metrics. The study supports the reduction of carbon emissions and the improvement of clean energy access in order to contribute to sustainable development goals.

In the study of Wang, et al. [15], Modified Differential Evolution (DE) algorithm and RF Regression were used to optimize renewable energy production in smart building micro grids. Multi-objective optimization problems, in which economic and ecological factors are taken into account, were solved with DE-based artificial intelligence techniques; power exchange constraints and energy costs were taken into account. In addition, digital twin models of renewable energy sources (solar panels and wind turbines) were created, their behaviors in different environmental conditions were simulated and a comprehensive approach was developed for energy management. The study verified the effectiveness of the method by comparing the performance of the DE algorithm with Particle Swarm Optimization (PSO).

In the study of Piper, et al. [16], a mixed reality-based system was designed for the visualization and management of street lighting systems of smart cities using digital twin technology. Real-time measured model street lamp data and real lighting data of the city of Durham were used, and machine learning was applied to estimate electricity purchase efficiency and adaptive brightness levels. The application developed with Unity Pro was run on Microsoft HoloLens 2 and allowed the user to view information such as voltage, current, carbon emission, electricity price, battery charge status, and LED mode, and to control the model street lamp. While electricity purchase was planned at appropriate times with the Deep Q-Network algorithm, the LSTM algorithm successfully predicted traffic flow with 12% RMSE and 20% MAPE error rates.

In the study of Tariq, et al. [17], the integration of phase change materials (PCM) was optimized with digital twin technology to reduce carbon emissions in the building sector. By performing simulations in four climatic zones, ventilation rate (ACH) and energy efficiency were estimated with high accuracy using multi-layer perceptron (ANNs). In the sensitivity analysis, solar radiation and solar chimney width were determined as the most effective parameters. The results showed that energy efficiency between 37%–40% and ventilation of 2.19–4.9 ACH could be achieved in different climates. The study emphasizes that digital twin is effective in optimizing building energy performance.

In their study, El-Gohary, et al. [18] designed an (ANN) based digital twin model to examine the energy consumption of residential buildings in Lebanon. The aim is to estimate energy consumption by evaluating different design alternatives of building materials such as exterior walls, roof insulation and windows, thus contributing to green building design. The digital twin model was developed using 1540 different insulation thickness, conductivity values and window type simulation data with eQuest software. The results showed that the ANN model agreed with the simulation outputs with high accuracy.

Khan, et al. [19] aim to examine how digitalization tools, including supply chain digitization, non-intrusive load monitoring, and digital twin technologies, enhance energy efficiency policy cycles in 30 Asian developing countries. The study finds that these technologies improve resource management and economic outcomes, helping policymakers set and achieve sustainable energy targets, although supply chain digitization may also have some negative side effects.

Bayer and Pruckner [20] investigated how replacing gas boilers with heat pumps affects CO2 emissions using a digital twin of all residential buildings in Germany. It was found that the application of heat pumps as a sole heating system reduces annual average CO2 emissions by 18.6% to 38.9%. The highest emission reductions were achieved by combining heat pumps with solar panels and battery systems, which was between 31.2% and 43.6% in buildings. From an economic perspective, it was concluded that the installation of heat pumps and PV is the most advantageous solution in terms of cost and emission reduction for most buildings.

In the study of Zhao, et al. [21] a building energy model (BEM) was created using BIM and 3D laser scanning technology in the evaluation of renovation plans of existing buildings. The aim is to increase the energy efficiency of existing buildings within the framework of the concept of nearly zero energy buildings (nZEBs) and to meet the energy demand with the use of clean energy. According to the case study, the proposed nZEB solution reduces building energy costs by 14.1%, increases solar PV energy production by 24.13% and reduces annual carbon emissions by approximately 4306 kg CO2 equivalent.

Wang, et al. [22] aimed to develop a highly accurate energy-consumption forecasting method for laser melting manufacturing by integrating fusionable transfer learning with digital-twin-based representations of fabrication processes. Their proposed FTL framework effectively predicts energy use across different product scales—achieving errors below 0.83%—and supports cost estimation and carbon-neutral manufacturing strategies.

Schmidt, et al. [23] aimed to evaluate pathways toward climate neutrality in the manufacturing and chemical–pharmaceutical industries by comparing Germany's situation with international experiences, particularly from Asia-Pacific and Arabic countries. Their analysis shows that existing German studies are largely theoretical, lack integrated economic evaluation, and indicate that green manufacturing technologies remain up to three times more expensive. To make climate-neutral production feasible, the authors highlight the need to accelerate research on more efficient and economically viable solutions—such as reaction intensification, catalyst optimization, sector coupling, and the integration of digital twins and process analytical technologies—to reduce costs and enable highly automated, sustainable manufacturing systems.

While our methodological framework relies on three machine learning (ML) models— RF, XGBoost, and LSTM—there has been a parallel rise in newly proposed metaheuristic optimization algorithms that contribute to solving high-dimensional and multi-objective problems in the broader energy optimization literature. Several recently introduced metaheuristic optimization algorithms, which reflect current developments in the field, are summarized below.

The Fashion Designer Algorithm (FDA) is a novel, parameter-free metaheuristic inspired by adaptive strategies in fashion design, which demonstrates superior convergence, robustness, and stability across a wide range of complex benchmark optimization problems [24]. The Detective Algorithm (DA) is a parameter-free, detective-inspired metaheuristic that uses a two-phase search strategy to achieve superior performance, stability, and global-optimum convergence across a wide range of complex, real-world constrained optimization problems [25]. The Search and Rescue Algorithm (SRA) is a novel, parameter-free metaheuristic inspired by real-world search and rescue operations, demonstrating robust, accurate, and stable performance across complex constrained engineering optimization problems [26]. The Antarctic Ice Worm Algorithm (AIWA) is a nature-inspired, two-phase metaheuristic that models the survival and foraging behaviors

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of Antarctic ice worms, achieving robust, consistent, and top-ranked performance across a wide range of benchmark optimization problems [27]. The Key Maker Algorithm (KMA) is a population-based, locksmith-inspired metaheuristic that balances exploration and exploitation to achieve high accuracy, reliability, and superior performance across diverse constrained engineering optimization problems [28]. The Psychologist Algorithm (PA) is a psychology-inspired metaheuristic that models candidate solutions as “clients” with distinct behavioral types, achieving superior stability, reliability, and near-optimal performance across complex engineering optimization problems [29]. The Library and Readers Algorithm (LRA) is a human-inspired, parameter-free metaheuristic that models readers’ selective search behavior to achieve superior accuracy, robustness, and convergence performance across a wide range of benchmark optimization problems [30]. The Driver and Navigator Algorithm (DNA) is a population-based, cooperative metaheuristic that efficiently balances global exploration and local exploitation, consistently outperforming state-of-the-art algorithms in solution quality, convergence speed, stability, and robustness across diverse benchmark optimization problems [31]. The Court and Judge Algorithm (CJA) is a population-based, judicial system–inspired metaheuristic that effectively balances global exploration and local exploitation, achieving superior robustness, stability, and solution quality across diverse constrained engineering optimization problems [32]. The Thunderstorm and Cloud Algorithm (TCA) is a parameter-free, weather-inspired metaheuristic that models cloud, lightning, wind, and rainfall behaviors to achieve superior solution quality, convergence, and robustness across a wide range of complex benchmark optimization problems [33]. The Community-Based Crisis Management Algorithm (CCMA) is a parameter-free, behaviorally inspired metaheuristic that balances exploration and exploitation, demonstrating superior performance, stability, and adaptability across high-dimensional, nonlinear, and constrained benchmark optimization problems [34]. The Perfumer Optimization Algorithm (POA) is a perfume-inspired, two-phase metaheuristic that effectively balances exploration and exploitation, achieving superior performance in solving complex, constrained routing optimization problems in transportation networks [35]. The Salamander Optimization Algorithm (SOA) is a nature-inspired, parameter-free metaheuristic that models salamanders’ adaptive and regenerative behaviors, achieving superior efficiency, robustness, and convergence across diverse complex benchmark optimization problems [36].

The studies are summarized in Table 1, and in the subsequent Discussion section, the results of this work are analyzed and compared with those reported in the literature.

TABLE 1. Summary of literature studies.

Author(s)	Method / Digital Twin Use
Park, et al. [9]	Solar Hydrail Digital Twin
Castilla, et al. [10]	Neural Network Digital Twin
Mohamad Zaidi, et al. [11]	City-Scale Digital Twin
Mingorance, et al. [12]	Reinforcement Learning Digital Twin
Zhao, et al. [13]	BIM-Based Digital Twin
Vilakazi and van Zyl [14]	Solar PV Digital Twin
Wang, et al. [15]	Differential Evolution–RF Digital Twin
Piper, et al. [16]	Mixed Reality Digital Twin
Tariq, et al. [17]	Phase-Change Material Digital Twin
El-Gohary, et al. [18]	ANN-Based Digital Twin
Khan, et al. [19]	Policy-Oriented Digital Twin
Bayer and Pruckner [20]	Residential Building Digital Twin
Zhao, et al. [21]	BIM-Energy Digital Twin
Wang, et al. [22]	Transfer Learning Digital Twin
Schmidt, et al. [23]	Industrial Process Digital Twin

The present study introduces a novel, AI-augmented digital twin architecture that extends beyond conventional forecasting by integrating real-time sustainability analytics and dynamic product-level pricing. Unlike prior works that model specific components or rely solely on simulation environments in their digital twin application for solar-powered vehicles. This research is grounded in actual industrial operations and real data flows from a vertically integrated manufacturing facility. By uniting energy forecasting, carbon footprint quantification, and pricing optimization within a single scalable framework, this study positions digital twin technology not merely as a monitoring tool, but as a strategic decision-making engine for climate-conscious, cost-efficient textile production

This study aims to develop and validate an integrated digital twin framework that enables solar energy forecasting, carbon footprint accounting, and AI-powered dynamic pricing in the textile manufacturing sector. By leveraging real-time operational data from a real-world facility, the model demonstrates how Industry 4.0 tools can unlock climate-smart decision-making.

The research focuses on a vertically integrated textile manufacturer—ATLAS Denim Co.—operating in Adana, Türkiye. It includes ML-based solar forecasting, carbon cost estimation, product-level sustainability reporting, and dynamic pricing mechanisms informed by environmental indicators.

Despite the growing body of work on digital twins and renewable energy integration, most existing studies remain either simulation-oriented or focused on isolated components, such as single processes, buildings, or vehicle systems. Only a limited number of contributions address factory-wide, real-time integration of PV forecasting, carbon footprint allocation, and pricing mechanisms within a single framework, and applications in small- and medium-scale textile plants using real operational data are particularly scarce. Furthermore, while carbon accounting and dynamic pricing are widely discussed conceptually, their joint implementation using machine learning-based forecasts in an industrial digital twin context has not been thoroughly explored.

To address this gap, the present study proposes and validates an AI-augmented digital twin framework for a vertically integrated textile manufacturer. The framework links solar energy forecasting, product-level carbon footprint estimation, and dynamic pricing into a unified decision-support system. The main contributions of this work are as follows:

- An integrated digital twin architecture is developed that simultaneously handles PV energy forecasting, carbon footprint reporting, and dynamic product-level pricing for textile manufacturing.
- A real-world industrial case study is presented at ATLAS Denim Co. in Adana, Türkiye, using real-time and historical operational data from an on-site rooftop PV system and production processes.
- A comparative evaluation of three ML models (RF, XGBoost, and LSTM) for solar energy forecasting is performed, demonstrating that the RF model achieves the best trade-off between accuracy and interpretability.
- A carbon-aware costing and pricing framework is introduced that allocates energy and emission costs at the product level and incorporates them into a dynamic pricing formula, supporting compliance with emerging ESG and regulatory requirements.
- Scenario and sensitivity analyses are conducted to demonstrate how the proposed framework can serve as a decision-making engine for climate-conscious and cost-efficient production planning in energy-intensive textile operations.

II. MATERIALS AND METHODS

A. DATA COLLECTION AND PREPROCESSING

To enable accurate solar energy forecasting and carbon footprint estimation at the product level, this study integrates multiple heterogeneous data sources encompassing environmental, operational, and energy-related parameters. The data collection and preprocessing phase was designed to ensure high-quality, consistent, and machine-learning-ready datasets.

1) DATA SOURCES AND TYPES

- Meteorological Data: External climate variables, including ambient temperature, solar radiation, humidity, wind speed, and cloud coverage, were retrieved from reputable online platforms such as WeatherSpark and Weather Atlas. These sources provided historical and real-time weather data with high temporal granularity.

- PV System Data: Electrical and physical measurements from the rooftop PV system of ATLAS Denim Co. were gathered using the system's integrated monitoring tools. The dataset included DC voltage and current, AC power output, panel temperature, and real-time solar irradiance.

- Production and Consumption Data: Facility-level operational metrics such as electricity consumption, shift-based production volumes, and machine operating schedules were collected from the company’s internal energy management and ERP systems.

2) TEMPORAL ALIGNMENT AND RESAMPLING

All data streams were synchronized using timestamp-based temporal alignment to match measurements from different sources on a common time axis. A fixed interval resampling method (e.g., 10-minute or hourly intervals) was applied to unify sampling frequencies and fill in temporal gaps.

3) MISSING DATA HANDLING

To address incomplete observations: Linear interpolation was applied for short-term gaps in continuous sensor data. Mean imputation was used for longer gaps or categorical-aggregated values. This ensured data continuity without distorting temporal trends.

4) OUTLIER DETECTION AND REMOVAL

Outliers—such as unrealistically high irradiance or negative power values—were identified and removed using z-score-based anomaly detection, where values beyond ±3 standard deviations from the mean were flagged as anomalies.

5) NORMALIZATION AND SCALING

To prepare the data for ML models: All numerical features were normalized to a [0, 1] range using min-max scaling, preserving relative relationships and accelerating training convergence. Categorical variables (if any) were one-hot encoded to maintain model compatibility.

6) DATA INTEGRATION AND MASTER FRAME FORMATION

After preprocessing, all datasets were merged into a unified master data frame, with consistent indexing and time

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alignment. This comprehensive frame formed the foundation for both forecasting and carbon-intensity analysis.

7) FEATURE SELECTION

A Pearson correlation analysis was conducted to quantify the linear relationship between input variables and the target variable (solar energy generation). Features with the highest correlation—such as solar irradiance, panel temperature, and ambient temperature—were selected as model predictors. This selection enhanced both the predictive power and computational efficiency of the modeling phase.

8) TIME-SERIES FORECASTING PIPELINE AND VALIDATION

To ensure a rigorous and reproducible time-series forecasting setup, the strictly chronological modelling pipeline was adopted. After the data collection and preprocessing steps described above, all time-indexed observations were ordered and split into training and test sets without any shuffling. Approximately 80% of the earliest observations were assigned to the training set, while the remaining 20% of the most recent observations were reserved for out-of-sample evaluation. This chronological split prevents information from the future from leaking into the training phase and reflects a realistic deployment scenario in which models are calibrated on past data and used to forecast unseen future conditions.

During model development, hyper parameter tuning was conducted exclusively within the training set using a rolling-origin cross-validation scheme, where the training window was progressively expanded and forecasts were generated for subsequent time blocks. The test set was never used for model selection, thus avoiding data leakage and optimistic bias in performance estimates.

To benchmark the proposed ML models, we implemented three baseline forecasting approaches:

- a persistence model, which assumes that the PV output at time t equals the last observed value;
- a classical ARIMA model calibrated on the univariate PV generation series; and
- a Prophet model, which is widely used for capturing trend and seasonality in energy time series.

These baselines provide interpretable reference points and allow us to quantify the added value of the RF, XGBoost, and LSTM models beyond standard time-series techniques.

By explicitly separating training, validation, and testing in time, and by including multiple baseline models, the risk of data leakage was minimized and provided a transparent assessment of model generalization performance in real-world deployment conditions."

B. MACHINE LEARNING METHODS

The generation of renewable energy—particularly from solar power systems—is inherently stochastic, nonlinear, and influenced by a variety of meteorological and environmental parameters. Factors such as solar irradiance, ambient temperature, humidity, and cloud cover interact in complex ways, creating challenges for traditional statistical forecasting models. These conventional approaches often assume stationarity, linearity, or independence, and thus fail to capture the temporal and multivariate intricacies inherent in solar energy datasets [37].

To address these limitations, this study adopts a supervised ML framework in which models learn historical patterns to forecast future energy output from PV systems. ML algorithms are particularly well-suited for this task due to their ability to:

- Model nonlinear and multivariate dependencies
- Handle high-dimensional feature spaces
- Adapt to non-stationary temporal trends and seasonality
- Resist overfitting through built-in regularization techniques

Three ML algorithms were selected based on their proven performance in renewable energy forecasting literature and their ability to generalize across time-series data: RF, XGBoost, LSTM. The integration of these three distinct yet complementary models allows for a comprehensive evaluation of forecasting performance across different learning paradigms—tree-based ensembles (RF, XGBoost) and deep learning architectures (LSTM). This diverse approach enhances model robustness, enabling:

- Short- and long-term forecasting
- Uncertainty-aware scenario generation
- Informed carbon footprint and cost analyses
- Integration into decision-support systems for green energy planning

By combining statistical rigor with algorithmic flexibility, this modeling strategy lays the groundwork for scalable, interpretable, and actionable forecasting systems in renewable energy management. A comparison of ML models is presented in Table 2 below.

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TABLE 2. The comparison of ML models.

Model	Model Type	Handles Nonlinearity	Time-Series Aware	Overfitting Resistance	Interpreta-bility	Training Complexity	Best Use Case
RF	Ensemble (Bagging)	Yes	Partial	High	High	Low	Baseline with robust performance and interpretability
XGBoost	Ensemble (Boosting)	Yes	Partial	High	Medium	Medium	High-accuracy forecasting with tuned performance
LSTM	Deep Learning (RNN)	Yes	Yes	Medium	Low	High	Capturing complex temporal patterns and long-term seasonality

1) RANDOM FOREST REGRESSOR

RF is an ensemble learning method comprising multiple decision trees. Each tree is trained on a bootstrap sample of the data and produces predictions that are averaged to form the final output. This approach can effectively model complex, nonlinear relationships across high-dimensional data spaces while mitigating overfitting through randomized feature selection and bagging [38]. Moreover, it provides intrinsic measures of feature importance, enhancing interpretability.

Due to its ability to capture nonlinear impacts of environmental factors—such as solar irradiance, temperature, and humidity—on energy production, it was included as a foundational baseline model in this research.

RF is an ensemble regression algorithm based on decision trees. Its key components can be expressed as follows:

- Bootstrap Sampling: Create B subsets from the training dataset using random sampling with replacement.

$$D_b = \{(x_i, y_i)\}, i = 1, \dots \dots, n_b; \quad b = 1, \dots . . B \tag{1}$$

- Train Decision Trees: For each subset D_b, train a regression tree $T_b(x)$. At each split node, select a random subset of features, and split using least squared error.
- Predict by Averaging: For a new input x, the final prediction is the average of all tree outputs:

$$\hat{y} = \frac{1}{B}\sum_{b=1}^{B} T_b(x) \tag{2}$$

The pseudo code of the RF Regressor predictor is given in Figure 1 below.

```
Algorithm: Random Forest Regression for Solar Energy Forecasting

Input:
- Training dataset D = {(x_1, y_1), (x_2, y_2), ..., (x_n, y_n)}
- Number of trees B

Output:
- Regression function ŷ = RF(x)

Procedure:
1. For b = 1 to B:
   a. Create a bootstrap sample D_b from D
   b. Train a regression tree T_b on D_b:
       i. At each node:
          - Select a random subset of features
          - Split on the feature that minimizes squared error

2. For a new input x:
   a. Compute predictions T_1(x), T_2(x), ..., T_B(x)
   b. Return ŷ = (1/B) * Σ_{b=1}^{B} T_b(x)
```

FIGURE 1. Pseudo code for RF Regressor.

2) XGBOOST

XGBoost builds on the gradient boosting framework, iteratively adding decision trees that correct errors made by previous ones. Known for its regularization, tree pruning, and parallel processing capabilities, XGBoost frequently achieves state-of-the-art accuracy in regression tasks, including those in renewable energy forecasting [39].

In our study, XGBoost was configured to improve upon RF by leveraging optimized learning rates, tree depth, and column subsampling, targeting more efficient convergence. It is particularly capable of modeling complex dependencies between meteorological predictors and energy output.

XGBoost is a powerful ensemble learning algorithm based on gradient boosting of decision trees. It builds models sequentially, where each new model attempts to correct the errors of its predecessor using gradient descent optimization.

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Given a training dataset:

$$D_b = \{(x_1, y_1), (x_2, y_2), \dots \dots .., (x_n, y_n)\} \tag{3}$$

The XGBoost model constructs an additive model of K trees:

$$\widehat{y_i} = \sum_{k=1}^{K} f_k(x_i), \quad f_k \in F \tag{4}$$

Where:

- F is the space of regression trees
- Each function f_k maps an input x_i to a predicted value at a tree leaf

The objective function optimized at each iteration is:

$$Obj(\theta) = \sum_{i=1}^{n} I(y_i, \widehat{y_i}) + \sum_{k=1}^{K} \Omega(f_k) \tag{5}$$

Where:

- ***I*** is differentiable loss function (e.g., squared error)
- The regularization term is as follows, penalizing model complexity (***T***= number of leaves, ***w***= leaf weights)

$$\Omega(f) = \gamma T + \frac{1}{2}\lambda \; \|w\|^2 \tag{6}$$

The pseudo code of the XGBoost predictor is given in Figure 2 below. The algorithm iteratively fits regression trees using first- and second-order gradients and updates predictions at each boosting round to construct the final model.

```
Algorithm: XGBoost Regression for Solar Energy Forecasting

Input:
- Training dataset D = {(x₁, y₁), (x₂, y₂), ..., (xₙ, yₙ)}
- Number of boosting rounds K

Output:
- Regression function ŷ(x) = Σ_{k=1}^{K} f_k(x)

Procedure:
1. Initialize predictions: ŷ_i = 0 for all i

2. For k = 1 to K:
   a. Compute gradients:
        g_i = ∂l(y_i, ŷ_i)/∂ŷ_i
        h_i = ∂²l(y_i, ŷ_i)/∂ŷ_i²
   b. Fit regression tree f_k(x) to minimize:
        Σ (g_i * f_k(x_i) + 0.5 * h_i * f_k(x_i)^2) + Ω(f_k)
   c. Update predictions:
        ŷ_i ← ŷ_i + f_k(x_i)

3. Return final prediction function:
    ŷ(x) = Σ_{k=1}^{K} f_k(x)
```

FIGURE 2. Pseudo code for XGBoost.

3) LSTM (LONG SHORT-TERM MEMORY)

LSTM networks are specialized RNN architectures designed to learn long-term dependencies in sequential data. By utilizing gated memory cells, they overcome the vanishing gradient issue inherent in classical RNNs, making them suitable for long-horizon forecasting in energy systems [40].

In this research, the LSTM model ingests historical PV generation data along with synchronous environmental time series (irradiance, temperature). This enables it to model long-term temporal dependencies and interactions, producing projections for future production horizons.

LSTM is a type of RNN designed to handle time series and sequential data. It solves the vanishing gradient problem seen in traditional RNNs by introducing memory cells and gating mechanisms.

At each tine step t, the LSTM cell performs the following operations:

Gate and State Equations:

- Forget gate:

$$f_t = \sigma(W_f \,.\, [h_{t-1}, x_t] + b_f) \tag{7}$$

- Input gate:

$$i_t = \sigma(W_i \,.\, [h_{t-1}, x_t] + b_i) \tag{8}$$

- Candidate memory content:

$$\overline{c_t} = tanh(W_c \,.\, [h_{t-1}, x_t] + b_c) \tag{9}$$

- Cell state update:

$$c_t = f_t \,.\, c_{t-1} + i_t \,.\, \bar{c_t} \tag{10}$$

- Output gate:

$$o_t = \sigma(W_o \,.\, [h_{t-1}, x_t] + b_o) \tag{11}$$

- Hidden state (output):

$$h_t = o_t \,.\, tanh(c_t) \tag{12}$$

Where:

- x_t: input at time t
- h_{t-1}: hidden state from previous time step
- σ : sigmoid activation
- Tanh: hyperbolic tangent
- W, b: trainable weight matrices and bias vectors

The pseudo code of the LSTM predictor is given in Figure 3 below. This figure outlines the LSTM regression workflow used for solar energy forecasting. The process includes preprocessing the time-series data, defining an LSTM-based neural network, compiling the model with MSE loss and the Adam optimizer, training through backpropagation through time, and finally predicting future solar energy values.

```
Algorithm: LSTM Regression for Solar Energy Forecasting

Input:
- Time series data X = {x₁, x₂, ..., x_T}
- LSTM hyperparameters (units, layers, epochs)

Output:
- Predicted values ŷ

Procedure:

1. Preprocess:
   a. Normalize the input time series data
   b. Reshape data into supervised learning format (X_t → y_t)

2. Define Model:
   a. Input layer: (timesteps, features)
   b. One or more LSTM layers
   c. Dense output layer

3. Compile Model:
   - Loss function: Mean Squared Error (MSE)
   - Optimizer: Adam

4. Train:
   - Use training data to optimize weights via BPTT (Backpropagation Through Time)

5. Predict:
   - Feed future input sequences to trained model
   - Output predicted solar energy values ŷ
```

FIGURE 3. Pseudo code for LSTM.

III. FINDINGS

A. MODEL TRAINING PERFORMANCE COMPARISON

Understanding the linear relationships among input variables is a critical prerequisite for both feature selection and model interpretability within any data-driven modeling pipeline. To this end, a Pearson correlation matrix was constructed to evaluate the pairwise associations between solar energy production and various meteorological indicators. This matrix serves as a diagnostic tool for uncovering both meaningful predictive relationships and potential multicollinearity issues, particularly relevant for models sensitive to redundant or collinear inputs.

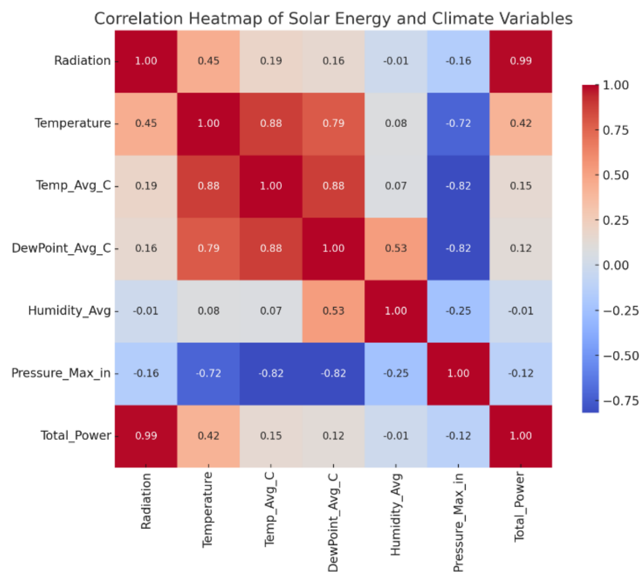

FIGURE 4. Correlation heatmap of solar energy and climate variables.

The resulting correlation structure is visualized through a heat map in Figure 4, which provides an intuitive and interpretable representation of the magnitude and direction of associations between variables. Such visualization enhances the transparency of the feature engineering process and grounds input selection in empirical evidence. From a modeling standpoint, correlation analysis facilitates:

- The identification of dominant predictive drivers,
- The elimination of irrelevant or noisy variables,
- And the mitigation of collinearity-induced performance degradation in models that assume feature independence (e.g., linear regression, gradient-boosted trees).

Key Observations from the Correlation Matrix (Figure 4) are as follows:

- Radiation vs. Total Power:

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A very strong positive correlation was observed between solar radiation and energy output, with a Pearson coefficient of $r \approx 0.99$. This result is consistent with the physical principles of PV systems and validates radiation as the primary explanatory variable for energy generation.

- Temperature vs. Total Power:

A moderate positive correlation was found ($r \approx 0.42$), indicating that higher ambient temperatures generally coincide with increased solar productivity. This is attributed to longer sunshine durations and clear-sky conditions typical of warmer days—especially relevant in Adana's subtropical climate.

- Pressure (Pressure Max in) vs. Total Power:

A slight negative correlation ($r \approx -0.12$) suggests an inverse relationship between atmospheric pressure and solar irradiance. This aligns with meteorological insights where high-pressure systems may be associated with cloud cover and reduced radiation.

- Humidity Average vs. Total Power:

No significant correlation was identified between average humidity and energy production, suggesting that humidity—at least within the operational thresholds observed—does not exert a meaningful impact on PV performance in this specific climatic context.

- Implications for Feature Engineering and Model Design

Based on the above findings, solar radiation and temperature were retained as primary predictive features in the subsequent ML models. This selection not only improves model performance through the inclusion of high-signal inputs but also ensures that the data pipeline is environmentally meaningful and contextually grounded.

Furthermore, the incorporation of domain-specific interpretation—such as the alignment of high temperature–irradiance correlation with Adana's regional climatology—enhances the scientific rigor and practical reliability of the analysis. By bridging data science techniques with climatological knowledge, the study achieves a more robust, generalizable, and explainable forecasting framework.

- Contribution to Sustainable Forecasting Systems

Finally, embedding correlation analysis into the feature selection process contributes to the sustainability-oriented architecture of the predictive system. It enables the development of models that are not only accurate and data-driven but also transparent, interpretable, and aligned with climate-relevant variables—a critical requirement for industrial applications such as carbon-aware decision-making in textile manufacturing.

1. RANDOM FOREST REGRESSOR

Figure 5 presents a comparative visualization of the observed versus predicted monthly PV energy production as estimated by the RF regression model over a one-year period. The predicted values closely track the observed production trends, underscoring the model's capacity to learn and replicate seasonal fluctuations inherent in solar energy generation.

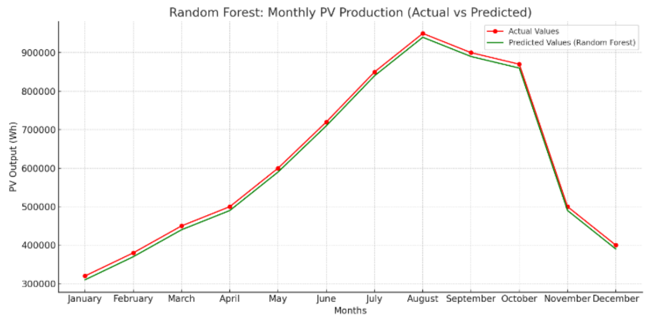

FIGURE 5. Monthly PV energy production: Actual vs predicted (RF Model).

Notably, the model demonstrates exceptional performance during high-radiation months—particularly between June and September—which coincide with peak solar irradiance conditions in southern Türkiye, a region characterized by strong solar potential. This alignment reinforces the model's suitability for climate-sensitive energy forecasting.

a) Nonlinear Modeling and Feature Sensitivity

The RF algorithm's ability to model complex, nonlinear relationships between key meteorological drivers—primarily solar radiation and ambient temperature—and PV energy output, is a significant strength. The ensemble learning approach, which integrates multiple decision trees through bootstrap aggregation (bagging), contributes to both its high accuracy and robustness against overfitting.

b) Quantitative Performance Metrics

The RF model achieved superior predictive accuracy among all models evaluated in this study, as reflected in the following performance indicators:

- R^2 (Coefficient of Determination): 0.993
- MAE: 350 Wh
- RMSE: 480 Wh

These metrics highlight the model's ability to provide reliable and low-variance forecasts, with minimal deviation from actual energy production values. The high R^2 value in particular indicates that over 99% of the variance in energy production can be explained by the model inputs.

c) Model Interpretability and Feature Importance

An analysis of the model's internal feature importance rankings further validates the robustness of its predictive logic. Solar irradiance and ambient temperature emerged as the most influential predictors—findings that are fully consistent with both domain knowledge and the results of prior correlation

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analysis. This enhances the interpretability of the model, which is essential for its integration into decision-support frameworks in industrial applications.

d) Strategic Relevance and Applicability

Due to its low error rates, resistance to overfitting, and strong generalization capacity, the RF Regressor was selected as the reference model for this study. Its effectiveness not only in capturing seasonal dynamics but also in maintaining predictive stability across all twelve months makes it particularly well-suited for:

- Dynamic energy pricing mechanisms
- Carbon footprint–sensitive production scheduling
- Real-time decision-making tools within digital twin architectures

The findings underscore the broader utility of ensemble learning techniques in enhancing the precision and reliability of forecasting systems within the renewable energy domain, especially in the context of industrial-scale applications such as sustainable textile manufacturing.

2. XGBOOST

Figure 6 illustrates the comparative analysis of observed versus predicted monthly PV energy production as estimated by the XGBoost regression model. The predictions demonstrate a high degree of alignment with actual production values across the calendar year, particularly during the mid-year period (May–September) when solar irradiance levels peak in the southern Türkiye region.

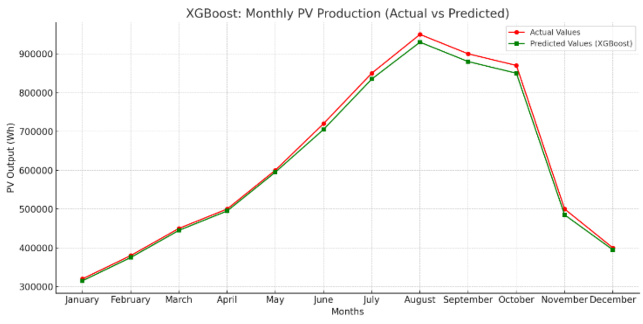

FIGURE 6. Monthly PV energy production: Actual vs predicted (XGBoost Model).

Despite minor underestimations during high-radiation months, such as August, the model effectively captures seasonal dynamics and nonlinear interactions among environmental variables, validating its appropriateness for renewable energy forecasting.

a) Algorithmic Strength and Learning Capacity

As an ensemble method based on gradient boosting, XGBoost constructs an additive sequence of decision trees where each subsequent tree corrects the residual errors of its predecessors. This iterative optimization mechanism, coupled with regularization and shrinkage techniques, enables the model to learn complex, nonlinear relationships with high precision and stability.

b) Quantitative Performance Metrics

The XGBoost model achieved high predictive accuracy, as summarized by the following statistical metrics:

- R^2 : 0.987
- MAE: 420 Wh
- RMSE: 530 Wh

Although these values are slightly less favorable than those obtained by the RF model, they still indicate strong model reliability and low residual error, especially considering the inherent volatility in solar energy datasets.

c) Computational Efficiency and Model Flexibility

One of the key advantages of XGBoost lies in its computational efficiency. The model exhibited faster training and inference times compared to RF, making it particularly attractive for real-time applications and edge computing environments. Furthermore, the algorithm offers greater flexibility in hyperparameter tuning, allowing for performance optimization under various operational constraints.

d) Generalization and Practical Relevance

XGBoost demonstrated robust generalization capabilities across varying ambient and radiation conditions, indicating its potential to operate effectively under heterogeneous climatic scenarios. These properties render it suitable for:

- Dynamic load forecasting
- Carbon-adjusted production scheduling
- Real-time energy awareness within digital twin systems

e) Conclusion and Strategic Role

In summary, while XGBoost incurs marginally higher error metrics than RF, it provides a compelling balance between predictive performance, computational speed, and model adaptability. Its demonstrated ability to capture environmental complexity and deliver scalable predictions positions XGBoost as a valuable forecasting tool in sustainability-driven industrial decision systems.

By supporting accurate, fast, and interpretable predictions, the XGBoost model contributes to the development of digitally intelligent and climate-responsive production ecosystems—particularly in the context of industrial-scale textile manufacturing and energy planning.

3. LSTM

Figure 7 displays the monthly PV energy generation forecasted by the LSTM neural network model in comparison to the actual observed values. The results highlight the model's strong ability to capture seasonal trends, especially during periods of peak solar irradiance—notably in July and August, when production levels are historically elevated in solar-rich regions such as southern Türkiye.

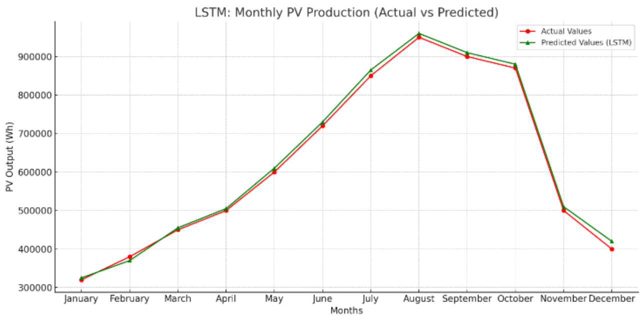

FIGURE 7. Monthly PV energy production: Actual vs predicted (LSTM Model).

a) Temporal Learning and Model Design

The LSTM architecture, a specialized variant of RNNs, is designed to learn long-term temporal dependencies by incorporating memory cells and gating mechanisms that regulate the flow of information across time steps. These gates—namely the input, forget, and output gates—allow the model to retain relevant information and filter out noise, making it particularly well-suited for time-series forecasting tasks involving weather and energy data.

In this study, the LSTM model was trained using sequences of historical PV output and corresponding meteorological variables, enabling it to infer temporal patterns that drive solar energy generation.

b) Performance Metrics

The LSTM model achieved the following predictive results:

- R^2 : 0.965
- MAE: 510 Wh
- RMSE: 600 Wh

These metrics indicate robust model performance, despite some overestimations observed in transition months such as March and November—periods characterized by fluctuating irradiance and variable atmospheric conditions. The relatively high R^2 score confirms the model's ability to generalize across a full year of energy production cycles.

c) Trade-Offs and Computational Considerations

While LSTM provides advanced modeling capabilities, it also introduces several computational and operational trade-offs:

- Training Complexity: The model requires extensive training time and significant computational resources, particularly as the number of layers, memory units, and sequence lengths increase.
- Hyper parameter Sensitivity: Model performance is highly dependent on proper tuning of hyperparameters such as learning rate, batch size, number of epochs, and look-back window size.
- Data Requirements: Deep learning models like LSTM generally necessitate larger and cleaner datasets to achieve optimal performance and avoid overfitting.

These considerations may pose challenges for real-time or edge-level deployment, especially in resource-constrained industrial environments.

d) Strategic Applicability and Value

Despite these limitations, the LSTM model offers substantial value in applications where temporal accuracy and pattern recognition are critical. Its ability to capture nonlinear sequential dynamics renders it particularly effective in the following contexts:

- Adaptive energy forecasting in smart factories
- Dynamic carbon footprint modeling
- Predictive maintenance and anomaly detection
- Digital twin–enabled decision support systems

By learning from historical dependencies in solar and meteorological data, LSTM contributes to the development of intelligent and responsive energy management infrastructures. Its application in this study enhances the forecasting ecosystem by introducing deep learning–driven capabilities that complement traditional ensemble models.

4. COMPARATIVE EVALUATION OF MODEL PERFORMANCE

To evaluate the forecasting performance of the ML models employed in this study, three widely accepted metrics were utilized: the R^2 Score, MAE, and RMSE. Among these, MAE is particularly critical, as it measures the average magnitude of prediction errors regardless of direction, providing a direct understanding of the model's practical accuracy. A lower MAE signifies a closer alignment between predicted and actual PV energy generation values, which is essential for reliability in real-world applications.

TABLE 3. A comparative summary of the three models.

Model	R^2 Score	MAE (Wh)	RMSE (Wh)
RF Regressor	0.993	350	480
XGBoost	0.987	420	530
LSTM	0.965	510	600

a) Interpretation of Results

As illustrated in Table 3, the RF Regressor achieved the highest R^2 value (0.993) and the lowest MAE (350 Wh), making it the most accurate model in this study. Its ability to effectively capture both linear and nonlinear interactions—particularly those between solar irradiance, ambient temperature, and energy output—enabled high-precision forecasts across varying seasonal conditions. Its consistent performance and resistance to overfitting make it highly suitable for industrial deployment.

The XGBoost model also delivered robust results, with a high R^2 score (0.987) and moderate error metrics. Thanks to its gradient boosting framework, XGBoost is particularly adept at handling complex feature interdependencies and nonlinearities. However, the slightly elevated MAE and RMSE compared to RF indicate a marginal trade-off between accuracy and adaptability, particularly under data variability.

The LSTM neural network, a deep learning model designed for sequential learning, demonstrated strength in modeling seasonality and lagged dependencies. Nevertheless, it recorded the highest error values among the three models (MAE: 510 Wh, RMSE: 600 Wh). The computational intensity, longer training times, and sensitivity to hyperparameters (e.g., learning rate, batch size, sequence length) may have impacted its stability across different time intervals and weather scenarios.

b) Model Recommendation

In addition to the three ML models, three baseline forecasters—a naïve persistence model, an ARIMA model, and a Prophet model—were evaluated. As summarized in Table 3, all three baselines were found to exhibit lower R^2 scores and higher MAE/RMSE values compared to the RF Regressor, indicating that the proposed approach provides a substantial accuracy improvement over standard time-series techniques.

To provide a more comprehensive view of model performance, the uncertainty of the RF forecasts was also quantified. For each prediction, the ensemble of trees generated a distribution of outputs from which 95% prediction intervals were derived. These intervals are shown as shaded bands around the predicted curves in Figures 5–7. The majority of the observed PV generation values fall within these bands, demonstrating that the model not only achieves low point-wise errors but also provides realistic bounds for planning under uncertainty. This is particularly important for industrial applications where underestimation or overestimation of solar output may lead to significant operational and economic risks.

Based on the comprehensive evaluation of forecasting accuracy and operational practicality, the RF Regressor emerges as the most suitable model for PV energy forecasting in this context. It offers a desirable combination of:

- High predictive performance,
- Interpretability and feature importance analysis,
- Low computational cost and rapid deployability.

These characteristics make RF a highly promising algorithm for integration into real-time digital twin platforms, carbon-aware production planning, and intelligent energy management systems—particularly within data-driven, sustainability-oriented industrial settings such as textile manufacturing.

B. ENERGY FORECASTING, CARBON FOOTPRINT ESTIMATION, AND DYNAMIC PRICING USING THE PROPOSED MODEL

In this study, a ML –based forecasting model was deployed to estimate the daily PV electricity generation of the on-site solar power plant (SPP) for the period between January and May 2025. The model utilized meteorological input features, specifically daily average temperature and solar radiation intensity, structured from historical climate datasets. Forecasts were generated on a day-by-day basis, enabling short-term energy planning.

The RF Regressor algorithm was selected as the core predictive model due to its demonstrated accuracy and interpretability in earlier stages of the research. Over the five-month period, the model forecasted a cumulative PV electricity generation of approximately 55.18 MWh. These predictions were subsequently integrated with production length data and electricity consumption metrics to calculate energy usage, carbon emissions, and energy cost per square meter of textile production.

TABLE 4. PV energy production.

Month	PV Energy Forecast (kWh)	Production (meters)	Energy (kWh/meter)	Carbon (kg/meter)	Energy Cost (₺/meter)
January	5,607.05	4,000,000	0.001402	0.000589	0.002804

February	7,775.26	4,200,000	0.001851	0.000778	0.003703
March	12,098.53	4,500,000	0.002689	0.001129	0.005377
April	13,087.59	4,800,000	0.002727	0.001145	0.005453
May	16,106.08	5,000,000	0.003221	0.001353	0.006442

As observed in Table 4, PV energy production increases progressively from January to May, consistent with rising levels of solar irradiance during spring and early summer. May stands out as the month with the highest PV yield, resulting in the greatest unit energy output, and by extension, the highest carbon savings and energy cost per meter of textile production.

To quantify environmental impact, the predicted energy generation was evaluated using a carbon emission factor of 0.42 kg CO_2 per kWh. This yielded an estimated carbon offset of approximately 23.9 metric tons over the five-month period. This outcome reflects the environmental benefit of renewable energy integration and supports compliance with carbon accounting frameworks. Integration into Decision Support and Sustainability Planning is as follows:

- These results provide critical insights for:
- Carbon footprint accounting,
- Dynamic pricing mechanisms,
- Sustainability-driven production planning,
- And real-time energy optimization in industrial operations.

The model outputs are structured for seamless integration into digital twin architectures, enabling continuous monitoring and decision-making aligned with sustainability targets. When coupled with production scheduling systems, this forecasting approach facilitates carbon-informed operational decisions at the unit level—offering textile manufacturers both economic and environmental visibility.

C. PRICING SIMULATION

The pricing model developed in this study is designed to generate dynamic, product-specific price offers by incorporating energy consumption metrics and carbon footprint values. This framework enables manufacturers committed to sustainable production practices to transparently evaluate their carbon-related costs and optimize pricing strategies in alignment with environmental performance indicators.

1) MODEL STRUCTURE AND KEY PARAMETERS

The model calculates the total energy consumed during production for each product and determines the proportion of this energy derived from renewable sources, specifically PV electricity. For products benefiting from a higher share of PV energy, the effective electricity cost per unit output decreases, as less grid-based (and typically fossil-fuel-intensive) energy is used. This directly leads to:

- Lower associated carbon emissions,
- Reduced carbon tax liability,
- And consequently, a more competitive unit price.

To link environmental impact with economic outcomes, the model incorporates both energy efficiency (in kWh/meter) and carbon intensity (in kg CO_2/meter), ensuring alignment with green production certification standards, particularly those required in international export markets.

$$\begin{aligned}&\mathit{Total\ Product\ Price}\\&=\mathit{Base\ Price}+\mathit{Energy\ Cost}+\mathit{Carbon\ Cost}\end{aligned} \tag{13}$$

Where:

$$\begin{aligned}&\mathit{Energy\ Cost}\\&=\left(\frac{\mathit{Energy\ Consum.}\ (kWh)}{\mathit{Length}\ (m)}\right) x\ \mathit{Electr.\ Unit\ Price}\end{aligned} \tag{14}$$

$$\mathit{Carbon\ Cost}=\left(\frac{\begin{array}{c}\mathit{PV\ Energy\ in\ Product.}\ (kWh)\\ x\ \mathit{Emission\ Factor}\ \left(\frac{kg\ CO_2}{kWh}\right)\end{array}}{\mathit{Length}\ (m)}\right) x\left(\frac{\mathit{Carbon\ Tax}}{1000}\right) \tag{15}$$

2) ML–ENHANCED DYNAMIC PRICING FRAMEWORK

The model is further enhanced by integrating ML -based PV energy forecasting, which refines the estimation of renewable energy contribution on a real-time basis. In this setup, the dynamic product price is defined as:

$$\begin{aligned}&\mathit{Dynamic\ Price}=\mathit{Base\ Price}+\mathit{Net\ Energy\ Cost}+\\&\mathit{Net\ Carbon\ Tax}\end{aligned} \tag{16}$$

Where:

$$\begin{aligned}&\mathit{Net\ Energy\ Cost}\\&=\left(\frac{E_{total}-E_{PV}^{ML}}{\mathit{Length}}\right) x\ \mathit{Electricity\ Unit\ Price}\end{aligned} \tag{17}$$

$$
\begin{aligned}
&Net\,Carbon\,Tax \\
&= \left(\frac{(E_{total} - E_{PV}^{ML})x\ EF}{Length}\right) x \left(\frac{V_{CO_2}}{1000}\right) \qquad (18)
\end{aligned}
$$

Here,

$\boldsymbol{E_{total}}$: Total energy required in production (kWh)

$\boldsymbol{E_{PV}^{ML}}$: PV energy generation predicted by ML (kWh)

$\boldsymbol{EF}$: Emission factor $\left(\frac{\mathbf{kg\ CO_2}}{\mathbf{kWh}}\right)$

$\boldsymbol{V_{CO_2}}$: Carbon tax (€/ton CO_2)

3) SCENARIO AND SENSITIVITY ANALYSIS OF DYNAMIC PRICING

To further strengthen the carbon and pricing analysis, scenario-based simulations were conducted integrating the ML –enhanced PV energy forecasts. The total product price now dynamically reflects the base price, net energy cost (adjusted for PV generation), and net carbon tax. Different scenarios were evaluated to account for variable carbon tax rates, electricity price fluctuations, and PV penetration levels. The results demonstrate that higher PV utilization improves both economic (unit price) and environmental (carbon footprint) performance, highlighting the dual benefits of renewable energy integration. This extension ensures that the pricing model can adapt to varying market and regulatory conditions, enhancing its practical relevance and providing a robust, data-driven framework for eco-sensitive decision-making in textile production. This pricing model allows for:

- Real-time integration of renewable energy impact into cost calculations,
- Internalization of environmental externalities (e.g., carbon pricing),
- And the development of flexible, eco-sensitive pricing strategies.

By quantifying the carbon and energy cost per meter of production, firms can generate product-specific, environmentally optimized price offers, which contribute to:

- Enhanced sustainability performance,
- Compliance with carbon regulation frameworks (e.g., CBAM),
- And strengthened market competitiveness in environmentally conscious export markets.

In scenarios with high PV energy utilization, both economic (cost) and environmental (carbon) indicators improve simultaneously. This dual benefit supports the transition toward green manufacturing while offering a data-driven basis for intelligent pricing and digital twin–enabled decision-making in industrial settings.

IV. DISCUSSION

Recent literature demonstrates that digital twin technologies and advanced data-driven models are increasingly being adopted to support energy efficiency, renewable energy integration, and carbon-emission reduction across various sectors—including buildings, transportation, manufacturing, and smart cities. The reviewed studies show diverse implementations: neural network–based digital twins for solar collectors [10], city-scale digital twins for residential energy consumption [11], reinforcement-learning-enabled optimization in industrial operations [12], BIM-integrated digital twins for zero-carbon building renovations [13], and deep learning–assisted digital twins for solar energy forecasting [14]. These efforts collectively highlight the growing convergence of ML, digitalization, and sustainability-driven decision-making.

Compared with these studies, the present research contributes a complementary yet distinct perspective by focusing specifically on the textile manufacturing industry, a high-energy, high-emission sector that has been underrepresented in digital twin–oriented sustainability research. While many existing digital twin applications address building energy systems, transportation, or general industrial operations, few studies operationalize renewable energy forecasting directly into production-level pricing, carbon accounting, and real-time manufacturing decisions—a gap that this study addresses.

The ML–based PV forecasting framework developed in this study provides predictive accuracy comparable to or exceeding similar models in the literature. For example, while Vilakazi and van Zyl [14] successfully used LSTM and RNN models to estimate solar plant energy generation, the RF Regressor here achieved an exceptionally high forecasting accuracy (R^2 = 0.993), outperforming both LSTM and XGBoost in the same context. Moreover, in contrast to the general-purpose forecasting in previous works, this study integrates predictions directly into carbon savings estimation, product-level energy cost allocation, and dynamic sustainability-based pricing, offering operational insights rather than purely technical forecasting outputs.

The study also aligns with emerging research emphasizing data-driven decision support within industrial environments. Similar to Mingorance, et al. [12], who combined digital twins with reinforcement learning to optimize industrial operations, the current work demonstrates how predictive modeling can enhance real-time production planning and sustainability performance. However, by focusing on renewable energy usage and carbon-tax–based pricing, the proposed framework extends digital twin concepts into environmentally informed economic decision-making, which is less explored in the reviewed literature.

Furthermore, the incorporation of PV forecasting into a digital twin–enabled decision support structure complements the broader trend highlighted in studies such as Wang, et al. [22], which use digital representations of manufacturing processes to optimize energy consumption. Unlike these works, which apply complex transfer learning or operator-learning methods, the present study shows that relatively simpler ML models—when properly calibrated—can provide high accuracy and immediate industry applicability.

Finally, by quantifying real carbon savings (23.9 tons over five months) and demonstrating how renewable energy forecasting can directly shape pricing strategies, this research introduces a practical pathway for integrating sustainability metrics into day-to-day manufacturing economics. This positions the study within the emerging literature that emphasizes the need for cross-disciplinary methods combining digital twins, AI, and sustainability policy (Khan, et al. [19]; Schmidt, et al. [23]), while offering a concrete, sector-specific solution tailored to textile production.

V. CONCLUSION

This study developed and validated a ML–based forecasting framework for estimating PV electricity generation and integrating its outputs into sustainability-focused decision-making processes in the textile industry. Using meteorological data—specifically solar radiation and ambient temperature—the model provided accurate daily forecasts of PV energy production, supporting real-time energy management, carbon accounting, and dynamic product pricing strategies. Among the tested algorithms, the RF Regressor demonstrated superior performance ($R^2 = 0.993$, MAE = 350 Wh, RMSE = 480 Wh), outperforming XGBoost and LSTM models in predictive accuracy and stability. The strong alignment between predicted and observed values across all models confirmed the effectiveness of data-driven approaches in modeling non-linear dependencies within renewable energy systems. Correlation analysis further reinforced the scientific basis of the model by identifying solar irradiance and temperature as the most influential predictors of PV output. The forecasts were operationalized to estimate carbon emissions and determine product-specific energy usage and cost per meter of textile production. Over a five-month evaluation period, the model predicted a total PV generation of 55.18 MWh, leading to an estimated carbon savings of 23.9 metric tons when benchmarked against conventional electricity sources. These outputs were used to inform a dynamic pricing model that incorporates both real-time energy costs and carbon taxation, enabling the internalization of environmental externalities within production economics. The proposed pricing system offers a flexible, transparent tool for aligning economic decisions with sustainability goals by emphasizing renewable energy contributions and reducing fossil-based carbon intensity. Overall, the study demonstrates the practical applicability and strategic value of ML in enhancing climate-resilient manufacturing practices, enabling digital twin–enabled decision support, and advancing sustainability-aligned pricing policies. Future research may focus on extending the forecasting framework to longer timeframes, incorporating real-time sensor data, and integrating multi-source renewable energy systems to further improve model adaptability and decarbonization impact. Future work includes extending the forecasting framework to longer timeframes, incorporating real-time sensor and IoT data, integrating multi-source renewable energy systems, exploring ensemble and hybrid ML techniques, and evaluating the scalability of the framework across multiple textile production facilities to further enhance predictive performance, operational efficiency, and carbon reduction impact.

REFERENCES

[1] *The International Energy Agency (IEA)*. Available: https://www.iea.org/reports/world-energy-outlook-2022

[2] K. Farhana, K. Kadirgama, A. S. F. Mahamude, and M. T. Mica, "Energy consumption, environmental impact, and implementation of renewable energy resources in global textile industries: an overview towards circularity and sustainability," *Mater. Circular Econ.*, vol. 4, no. 1, p. 15, 2022.https://doi.org/10.1007/s42824-022-00059-1.

[3] J. Beier, S. Thiede, and C. Herrmann, "Energy flexibility of manufacturing systems for variable renewable energy supply integration: Real-time control method and simulation," *J. Clean. Prod.*, vol. 141, pp. 648-661, 2017.https://doi.org/10.1016/j.jclepro.2016.09.040.

[4] Z. Asgari and F. P. Rahimian, "Advanced virtual reality applications and intelligent agents for construction process optimisation and defect prevention," *Procedia Eng.*, vol. 196, pp. 1130-1137, 2017.https://doi.org/10.1016/j.proeng.2017.08.070.

[5] W. Yu, P. Patros, B. Young, E. Klinac, and T. G. Walmsley, "Energy digital twin technology for industrial energy management: Classification, challenges and future," *Renew. Sustain. Energy Rev.*, vol. 161, p. 112407, 2022.https://doi.org/10.1016/j.rser.2022.112407.

[6] E. O'Dwyer, I. Pan, R. Charlesworth, S. Butler, and N. Shah, "Integration of an energy management tool and digital twin for coordination and control of multi-vector smart energy systems," *Sustain. Cities Soc.*, vol. 62, p. 102412, 2020.https://doi.org/10.1016/j.scs.2020.102412.

[7] M. Schneider, "Nuclear Power in France," Beyond the Myth, Commissioned by the Greens-EFA Group in the European Parliament2008.

[8] K. Yılmaz, İ. Ö. Aksu, M. Göçken, and T. Demirdelen, "Sustainable textile manufacturing with revolutionizing textile dyeing: Deep learning-based, for energy efficiency and environmental-impact reduction, pioneering green practices for a sustainable future," *Sustainability*, vol. 16, no. 18, p. 8152, 2024.https://doi.org/10.3390/su16188152.

[9] B. Park, J. Song, D. Eom, J. Choi, S. J. Kim, and S. Park, "Digital twin-based design and techno-economic analysis of solar hydrail as future locomotive," *Int. J. Hydrogen Energy*, vol. 56, pp. 1216-1226, 2024.https://doi.org/10.1016/j.ijhydene.2023.12.080.

[10] M. Castilla, J. L. Redondo, A. Martínez, and J. Álvarez, "Artificial Neural Network-based digital twin for a flat plate

solar collector field," *Eng. Appl. Artif. Intell.*, vol. 133, p. 108387, 2024.https://doi.org/10.1016/j.engappai.2024.108387.

[11] N. H. Mohamad Zaidi, C. H. Lim, and H. Razali, "Mitigating the energy consumption and carbon emissions of a residential area in a tropical city using digital twin technology: A case study of Bertam, Penang," *Buildings*, vol. 14, no. 3, p. 638, 2024.https://doi.org/10.3390/buildings14030638.

[12] R. Mingorance, D. C. Pereira, J. Uribetxebarria, and U. Leturiondo, "A methodology leveraging digital twins to enhance the operational strategy of manufacturing plants in unexpected scenarios," *Results Eng.*, p. 106761, 2025.https://doi.org/10.1016/j.rineng.2025.106761.

[13] L. Zhao, H. Zhang, H. Wang, Y. Wang, and T. Li, "A digital-twin evaluation framework of zero carbon buildings for existing residential buildings based on scan-to-BIM," *Alex. Eng. J.*, vol. 124, pp. 204-213, 2025.https://doi.org/10.1016/j.aej.2025.03.042.

[14] N. Vilakazi and T. van Zyl, "Deep learning forecasting of photovoltaics output using digital twin data," in *Southern African Conference for Artificial Intelligence Research*, 2024, pp. 405-419: Springer.

[15] Q. Wang, Y. Yin, Y. Chen, and Y. Liu, "Carbon peak management strategies for achieving net-zero emissions in smart buildings: Advances and modeling in digital twin," *Sustain. Energy Technol. Assess.*, vol. 64, p. 103661, 2024.https://doi.org/10.1016/j.seta.2024.103661.

[16] W. Piper, H. Sun, and J. Jiang, "Digital twins for smart cities: case study and visualisation via mixed reality," in *2022 IEEE 96th Vehicular Technology Conference (VTC2022-Fall)*, 2022, pp. 1-5: IEEE.

[17] R. Tariq, C. Torres-Aguilar, N. A. Sheikh, T. Ahmad, J. Xamán, and A. Bassam, "Data engineering for digital twining and optimization of naturally ventilated solar façade with phase changing material under global projection scenarios," *Renew. Energy*, vol. 187, pp. 1184-1203, 2022.https://doi.org/10.1016/j.renene.2022.01.044.

[18] M. El-Gohary, R. El-Abed, and O. Omar, "Prediction of an efficient energy-consumption model for existing residential buildings in Lebanon using an Artificial Neural Network as a digital twin in the era of climate change," *Buildings*, vol. 13, no. 12, p. 3074, 2023.https://doi.org/10.3390/buildings13123074.

[19] M. Khan, S. Khan, and I. U. Rahman, "Supply-chain digitization and non-intrusive load monitoring with digital twins: The Impact of Energy Policies on Carbon Pricing and Renewable Energy Deployment," *Energy Econ.*, vol. 144, p. 108400, 2025.https://doi.org/10.1016/j.eneco.2025.108400.

[20] D. R. Bayer and M. Pruckner, "Data-driven heat pump retrofit analysis in residential buildings: Carbon emission reductions and economic viability," *Appl. Energy*, vol. 373, p. 123823, 2024.https://doi.org/10.1016/j.apenergy.2024.123823.

[21] L. Zhao, H. Zhang, Q. Wang, and H. Wang, "Digital-twin-based evaluation of nearly zero-energy building for existing buildings based on scan-to-BIM," *Adv. Civ. Eng.*, vol. 2021, no. 1, p. 6638897, 2021.https://doi.org/10.1155/2021/6638897.

[22] L. Wang, J. Xu, S. Zhang, J. Tan, S. Fei, X. Shi, J. Pang, and S. Luo, "Energy consumption forecasting for laser manufacturing of large artifacts based on fusionable transfer learning," *Vis. Comput. Ind. Biomed. Art*, vol. 7, no. 1, pp. 1-14, 2024.https://doi.org/10.1186/s42492-024-00178-3.

[23] A. Schmidt, D. Köster, and J. Strube, "Climate neutrality concepts for the German chemical–pharmaceutical industry," *Processes*, vol. 10, no. 3, p. 467, 2022.https://doi.org/10.3390/pr10030467.

[24] G. Bektemyssova, Z. Montazeri, M. Dehghani, I. K. Ibraheem, A. Smerat, O. P. Malik, A.-S. Madhloom, and M. A. Ahmed, "Fashion Designer Algorithm: A human-creativity inspired parameter-free metaheuristic for optimization tasks," *Int. J. Intell. Eng. Syst.*, vol. 18, no. 10, 2025.https://doi.org/10.22266/ijies2025.1130.50.

[25] A. Zraiqat, B. Batiha, M. Al Soudi, I. K. Ibraheem, A. Smerat, Z. Montazeri, M. Dehghani, F. Werner, A. A. M. M. AL-Salih, M. A. J. I. J. o. I. E. Ahmed10, and Systems, "Detective Algorithm: A Novel Human-inspired Metaheuristic for Solving Real-world Constrained Optimization Problems," vol. 18, no. 10, 2025.

[26] G. Bektemyssova, Z. Montazeri, M. Dehghani, I. K. Ibraheem, A. Smerat, F. Werner, A. A. M. M. AL-Salih, M. A. Ahmed, and K. Eguchi, "Search and Rescue Algorithm (SRA): A metaheuristic approach for efficient constrained engineering optimization," *Int. J. Intell. Eng. Syst. Synth. Biol.*, vol. 18, no. 10, 2025.https://doi.org/10.22266/ijies2025.1130.39

[27] G. Bektemyssova, Z. Montazeri, M. Dehghani, I. K. Ibraheem, A. Smerat, O. P. Malik, A.-S. Madhloom, and M. A. Ahmed, "Antarctic Ice Worm Algorithm: A novel nature-inspired metaheuristic for optimization tasks," *Int. J. Intell. Eng. Syst.*, vol. 18, no. 10, 2025.https://doi.org/10.22266/ijies2025.1130.21.

[28] A. Zraiqat, O. Al Sayyed, M. Al Soudi, A.-S. Madhloom, A. Smerat, Z. Montazeri, M. Dehghani, O. P. Malik, M. A. Ahmed, and I. K. Ibraheem, "Key Maker Algorithm: A novel human-based metaheuristic for constrained optimization," *Int. J. Intell. Eng. Syst. Synth. Biol.*, vol. 18, no. 9, 2025.https://doi.org/10.22266/ijies2025.1031.45.

[29] A. Zraiqat, B. Batiha, O. Al-Refai, A. A. M. M. AL-Salih, A. Smerat, Z. Montazeri, M. Dehghani, F. Werner, M. A. Ahmed, and I. K. Ibraheem10, "Psychologist Algorithm: A human-inspired metaheuristic for solving complex constrained optimization problems," *Int. J. Intell. Eng. Syst.*, 2025.https://doi.org/10.22266/ijies2025.1031.09.

[30] A. Zraiqat, B. Batiha, M. Al Soudi, A.-S. Madhloom, A. Smerat, Z. Montazeri, M. Dehghani, O. P. Malik, M. A. Ahmed, I. K. J. I. J. o. I. E. Ibraheem, and Systems, "Library and Readers Algorithm (LRA): A Novel Human-Inspired Parameter-Free Metaheuristic for Efficient Global Optimization," vol. 18, no. 9, 2025.

[31] A. Zraiqat, O. Al Sayyed, M. Al Soudi, A. A. M. M. AL-Salih, A. Smerat, Z. Montazeri, M. Dehghani, F. Werner, M. A. Ahmed, and I. K. Ibraheem10, "Driver and Navigator Algorithm: A novel parameter-free human-inspired metaheuristic for efficient global optimization," *Int. J. Intell. Eng. Syst. Synth. Biol.*, vol. 18, no. 9, pp. 555-569, 2025.https://doi.org/10.22266/ijies2025.1031.36

[32] A. Zraiqat, B. Batiha, O. Al-Refai, M. A. Ahmed, A. Smerat, Z. Montazeri, M. Dehghani, O. P. Malik, A. A. M. M. AL-Salih, and I. K. Ibraheem, "Court and Judge Algorithm (CJA): A novel human-inspired metaheuristic for engineering optimization," *Int. J. Intell. Eng. Syst. Synth. Biol.*, vol. 18, no. 8, 2025.https://doi.org/10.22266/ijies2025.0930.05.

[33] A. Zraiqat, B. Batiha, O. Al-Refai, A. A. M. M. AL-Salih, A. Smerat, Z. Montazeri, M. Dehghani, W. Aribowo, M. A. Ahmed, and I. K. Ibraheem, "Thunderstorm and Cloud Algorithm: A novel parameter-free metaheuristic inspired by atmospheric dynamics for complex optimization tasks," *Int. J. Intell. Eng. Syst. Synth. Biol.*, vol. 18, no. 8, 2025.https://doi.org/10.22266/ijies2025.0930.11.

[34] A. Zraiqat, B. Batiha, O. Al-Refai, A. A. M. M. AL-Salih, A. Smerat, Z. Montazeri, M. Dehghani, O. P. Malik, M. A. Ahmed, and I. K. Ibraheem, "Community-Based Crisis Management Algorithm (CCMA): A novel parameter-free metaheuristic for complex constrained optimization," *Int. J. Intell. Eng. Syst. Synth. Biol.*, vol. 18, no. 8, 2025.https://doi.org/10.22266/ijies2025.0930.37.

[35] T. Hamadneh, B. Batiha, O. Al-Refai, M. A. Ahmed, A. Smerat, Z. Montazeri, M. Dehghani, W. Aribowo, A. A. M. M. AL-Salih, and I. K. Ibraheem, "Enhancing transportation routing efficiency with the perfumer optimization algorithm: A comparative metaheuristic study," *Int. J. Intell. Eng. Syst. Synth. Biol.*, vol. 18, no. 8, 2025.https://doi.org/10.22266/ijies2025.0930.01.

[36] T. Hamadneh, B. Batiha, O. Al-Refai, I. K. Ibraheem, A. Smerat, F. Werner, Z. Montazeri, M. Dehghani, R. K. Jawad, and A. Al-Salih10, "Salamander optimization algorithm: a new bio-inspired approach for solving optimization problems," *Int. J. Intell. Eng. Syst.*, vol. 18, pp. 550-561, 2025.https://doi.org/10.22266/ijies2025.0831.35.

[37] S. Dargan, M. Kumar, M. R. Ayyagari, and G. Kumar, "A survey of deep learning and its applications: a new paradigm to machine learning," *Arch. Comput. Methods Eng.*, vol. 27, no. 4, pp. 1071-1092, 2020.https://doi.org/10.1007/s11831-019-09344-w.

[38] T. N. Rincy and R. Gupta, "Ensemble learning techniques and its efficiency in machine learning: A survey," in *2nd international conference on data, engineering and applications (IDEA)*, 2020, pp. 1-6: IEEE.

[39] H. Abbasimehr, R. Paki, and A. Bahrini, "A novel XGBoost-based featurization approach to forecast renewable energy consumption with deep learning models," *Sustain. Comput. Inform. Syst.*, vol. 38, p. 100863, 2023.https://doi.org/10.1016/j.suscom.2023.100863.

[40] H. Zheng, J. Yuan, and L. Chen, "Short-term load forecasting using EMD-LSTM neural networks with a Xgboost algorithm for feature importance evaluation," *Energies*, vol. 10, no. 8, p. 1168, 2017.https://doi.org/10.3390/en10081168.

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